

Cardoso — Sage Connection Setup

Wiring a freshly deployed site into Sage 300 for BAT Reconciliation.

This guide covers the one-time setup needed on every newly deployed Cardoso site to give it read-only access to the customer's Sage 300 database. Without this connection, the BAT Reconciliation module shows "no Sage" and the weekly credit-note totals stay blank.

The whole setup takes about **5 minutes**.

1. What this does

The BAT Reconciliation module compares supplier invoice totals (extracted via OCR) against credit-note batches posted in Sage 300. To do this it needs a read-only SQL Server connection to the customer's Sage database.

The connection itself is just a row in Cardoso's local SQLite `databaseconnection` table. The BAT module auto-detects it by matching `name LIKE '%sage%' AND status = 'active'`, and runs two hard-coded queries against it:

- `credit_notes_by_week` — pulls AP credit-note batches for a given week from `APIBC` / `APIBH` / `APIBD`, filtered to BAT vendors.
- `posted_invoices` — pulls posted BAT invoices from `APOBL` for matching against extracted invoice numbers.

Both queries are stored in [src/services/batReconciliation.js](#) and a reference copy is written into the connection row's `table_configs` field at seed time, for audit purposes.

2. Prerequisites

Before you start, you'll need:

- The Cardoso site already installed at `C:\Cardoso Customer App` and the `Cardoso` Windows service running.
- The customer's Sage 300 SQL Server credentials:
 - host (e.g. `SAGE-SQL01` or an IP)
 - database name (e.g. `BATCOMPDAT`)
 - SQL username (read-only is fine)
 - SQL password
- A Cardoso admin login for the new site.

If you don't have the Sage credentials, get them from the customer's IT contact before you start — the seed script creates the row with placeholders, but the connection won't activate until they're filled in.

3. Run the seed script

The seed script creates (or refreshes) a row called `Sage 300 (BAT Reconciliation)` in the local Cardoso database. It pre-populates the BAT-specific queries and sets `is_bat_only = 1` so the row is excluded from the generic sync engine.

Open PowerShell **as Administrator** and `cd` to the install directory:

```
cd "C:\Cardoso Customer App"
```

The bundled Node binary lives at `.\node\node.exe`. Run the seed script against it:

```
.\node\node.exe scripts\seed-sage-connection.js
```

Expected output:

Seeded Sage connection (id 7) with credit-notes query and is_bat_only=1.

Current Sage connection row:

```
{
  id: 7,
  name: 'Sage 300 (BAT Reconciliation)',
  host: 'FILL_IN_SAGE_HOST',
  database_name: 'FILL_IN_SAGE_DATABASE',
  username: 'FILL_IN_SAGE_USERNAME',
  status: 'inactive'
}
```

Fill in these placeholders via the Connections page (Edit → save):

```
host:      FILL_IN_SAGE_HOST
database:  FILL_IN_SAGE_DATABASE
username:  FILL_IN_SAGE_USERNAME
password:  (blank – set via Edit modal)
```

Then Test → Activate. BAT module will pick it up automatically.

Note: If a Sage row already exists, the script updates the `table_configs` metadata in place. It will **not** overwrite a `sync_query` the user has already customised. Safe to re-run.

Common errors

- **node : The term 'node' is not recognized...** — `node` isn't on PATH for that shell. Use the full path `.\node\node.exe` as above.
- **SqliteError: no such table: databaseconnection** — the app hasn't run yet on this site, so migrations haven't created the table. Start the `Cardoso` service once (it'll run all pending migrations) then re-run the seed.

4. Fill in credentials via the UI

Open the Cardoso UI on the new site and sign in as an admin.

1. Go to **Settings** → **Connections**.
2. Find the row called **Sage 300 (BAT Reconciliation)** — status will say *Inactive*.
3. Click **Edit**.
4. Fill in:
 - **Host** — Sage SQL Server hostname or IP
 - **Port** — usually `1433`
 - **Database** — Sage company database name

- **Username** — SQL Server login
- **Password** — SQL Server password
- **Use Encryption** — leave off unless the customer's SQL Server requires it.

5. Click **Save**.

The password is encrypted at rest using the app's encryption key — it never appears in plain text in the database or in the UI after save.

5. Test and activate

Still in **Settings** → **Connections**:

1. Click **Test** on the Sage row. You should see *Connection successful* and a row count from the test query.
2. If the test fails, check:
 - Firewall — does the Cardoso machine have TCP 1433 access to the Sage SQL Server?
 - Credentials — try them in SSMS from the Cardoso machine to rule out a typo.
 - Encryption — some hardened SQL Server installs require `Use Encryption = on`. Toggle it and re-test.
3. Once the test passes, click **Activate**. Status flips to *Active*.

That's it on the connection side. The BAT Reconciliation module now sees the connection and starts populating credit-note totals on its next refresh.

6. Verify BAT Reconciliation is using it

1. Go to **Reconciliation** in the main nav.
2. The weekly comparison panel ("H1 · Weeks 1–26" / "H2 · Weeks 27–53") should show `BAT` and `Credit Notes` totals side by side. If both columns have values, the connection is working.
3. The `LAST WEEK PAID` and `MISSING WEEKS` cards at the top of Reconciliation also depend on this connection. They turn green/red based on the Sage data.
4. If the page still says *Awaiting data* or the Credit Notes column is all dashes, click **Refresh Sage cache** (top right of the weekly panel) to force an immediate pull. Otherwise the cache refreshes on the scheduler tick.

If you see *Sage refresh failed (...)* in the cache footer, the connection is active but a query failed — open the System Log (Settings → System Log) to see the SQL error.

7. What lives where

For future debugging, here's the relevant surface area:

Concern	Location
Seed script	<code>scripts/seed-sage-connection.js</code>
Hard-coded BAT queries	<code>src/services/batReconciliation.js</code>
Connection auto-detect	<code>name LIKE '%sage%' AND status = 'active'</code>
Connection row	SQLite table <code>databaseconnection</code>
Cache refresh schedule	<code>src/scheduler.js</code> (skipped on Hub mode)

Connections UI	Settings → Connections
Per-user access toggle	can_access_reconciliation

The connection is **never** synced to the Hub. It's a per-site secret and the Hub doesn't talk to Sage directly.

Summary

1. `cd "C:\Cardoso Customer App"`
2. `.\node\node.exe scripts\seed-sage-connection.js`
3. Settings → Connections → Edit Sage row → fill host / db / user / pass
4. Test → Activate
5. Open Reconciliation, confirm Credit Notes column populates

If anything in step 5 doesn't work, check the System Log for the SQL error before re-running the seed — re-seeding rarely helps once the row exists.